Design Document

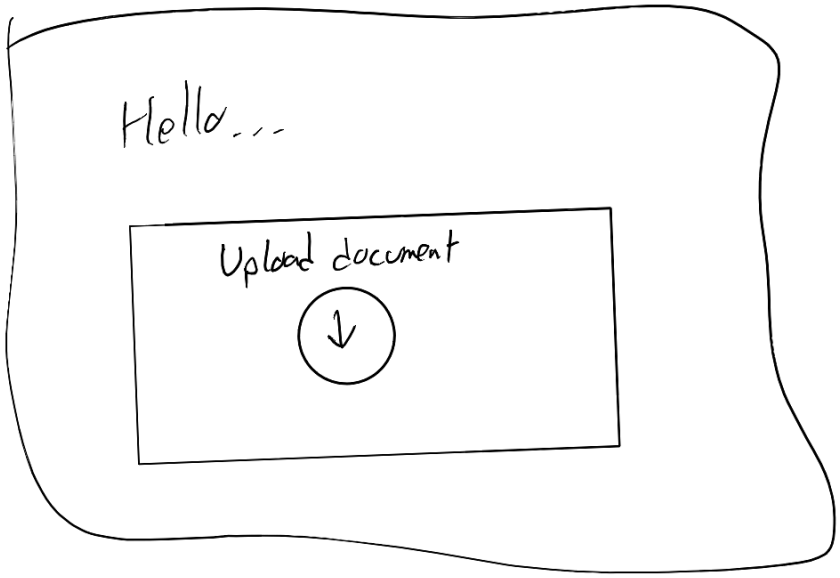
Design

User Flow Diagram

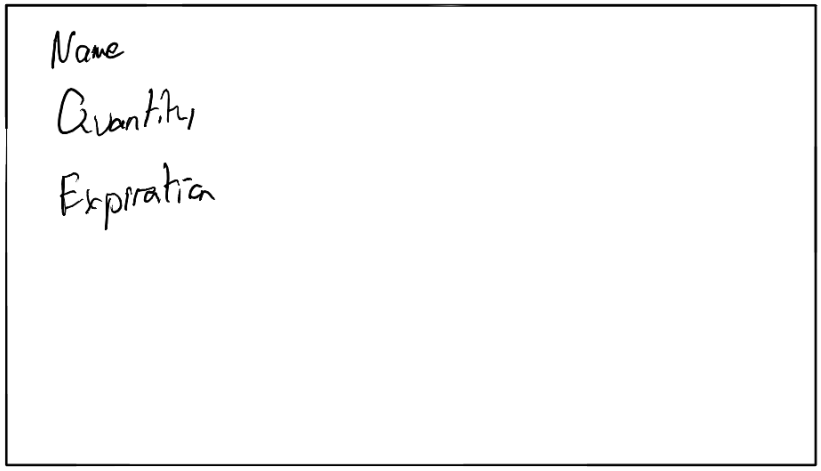
Initialize inventory amount
Taking picture of shelf
Manual input form
Generate initial list of inventory items
Confirm items
Edit items (if needed)
Overview Dashboard showing all item info
Item Click
Edit Name, Quantity, or Expiration Date
Edit Today's Usage Rate
Click Save
Enter Rate
New Value shows up immediately
Chart updates with new data point
Average Usage Rate is recalculated

Rough UI Mockups

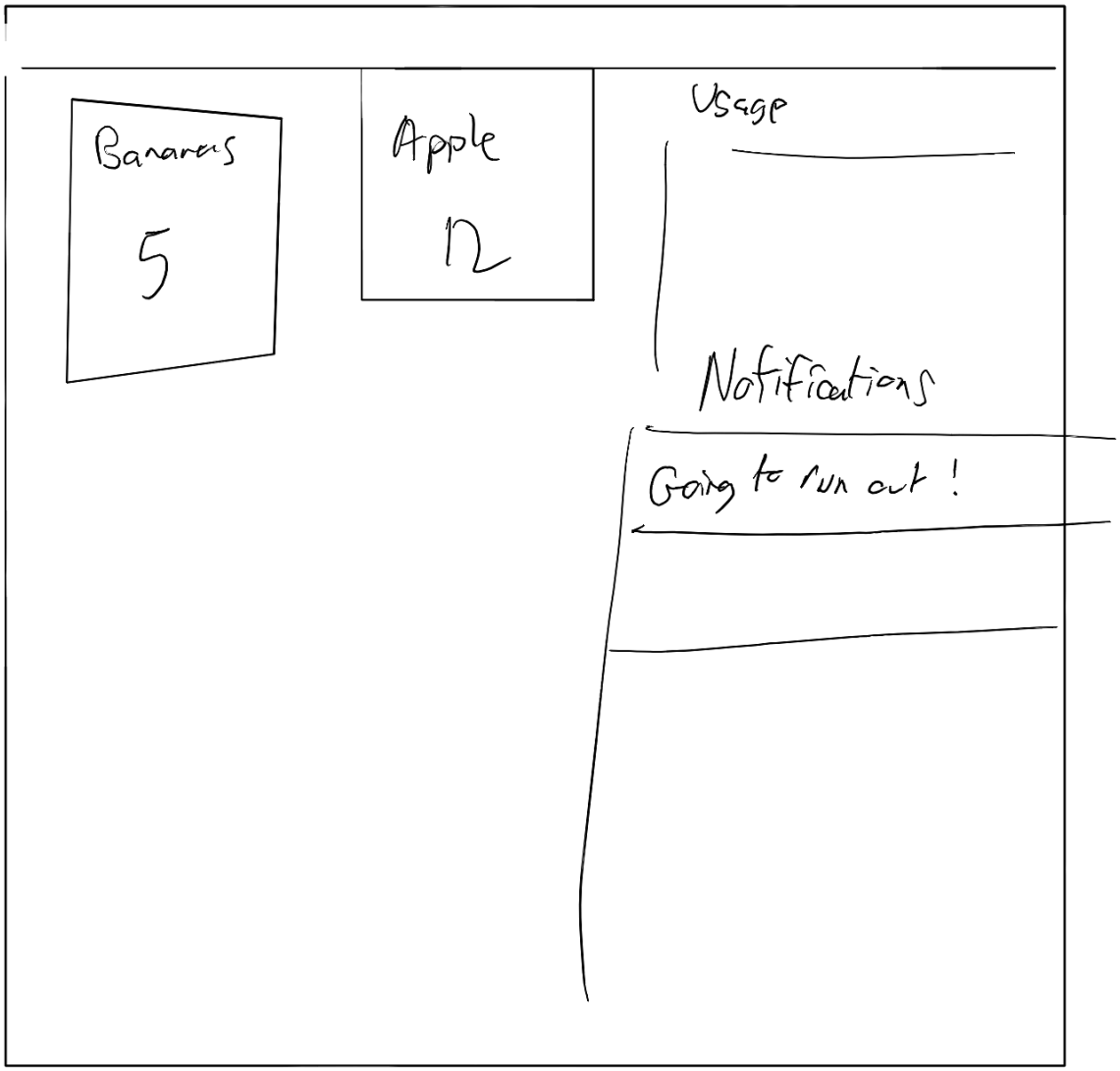
Initialize



Adding Item



Dashboard



Major requirements Brainstorm

- Track assets with AI
  - Figure out how low the inventory should get (percentage) before sending out a notification
  - Notifications sidebar
    - Notification scaling
- predict "shelf-life" or usage burnout
  - Need to figure out current usage and estimate
  - Trend diagram NEED
- Suggest sustainable procurement alternatives.—“Find greener, more ethical, long-term purchasing options.”
  - Least valuable option because it’s not part of the success metric

Final Features

- Testing with Playwright
- OpenAI integration for visual asset scanning
- Dashboard listing item quantity, expiration, and creation date
- Item editing (CRUD)
- Trend diagram
  - Randomly generated usage and data points that back up that usage trend
  - Allows user to see current usage for past 30 days and input existing usage
  - Flexibility to change
  - Average Usage rate: 1-20
  - Data points: 1-40

Tech Stack

- Next.JS is full stack application

Next.JS provides the development speed for the MVP, which is further compounded by my past experiences using it. The library removes the hassle of dealing with CORS issues from the frontend and backend running on two different ports. Additionally, having both the frontend and backend together allows for shared typing, making it easier to have consistency in how data is being processed throughout the application. The ecosystem around Next.JS provides out-of-the-box necessities such as TailwindCSS and React hot-reloading, granted me the ability to focus on the business logic rather than a lengthly environment setup.

- PostgreSQL for dockerized application

The use of the dockerized application was largely for consistency with the local development nature of the MVP. By using a PostgreSQL docker image, the MVP has room to expand to different possibilities such as cloud hosting or on-site servers hosting PostGreSQL images. The flexibility to expand means that a cost-effective solution can be researched instead of being locked-in to one provider. Additionally, PostgreSQL has support for richer data types, making it easier to use than having to worry about converting between primitive data types and richer data types.

- OpenAI for AI integration

Familiarity with the OpenAI API platform through previous projects allowed for the integration of the AI image recognition to be easier. I was aware through prior research that the OpenAI API platform had structured outputs, reducing the risk of receiving bad input from the AI.

- Playwright for testing

The Playwright library was an AI suggestion that was validated through web searches. The result of the web search was that Playwright was an end-to-end testing library that was directly mentioned in the Next.JS documentation here (https://nextjs.org/docs/app/guides/testing/playwright).

Future Enhancements

- Deployment to the cloud for greater availability compared to local deployment
- Barcode scanner
    - Barcode scanner would be first option, followed by AI image detection and last is manual entry
    - https://ericblade.github.io/quagga2/tutorials/react-integration.html
- Trend diagram showing trend over 30 day period
    - Should be able to expand view to months at a time (3 months, 6 months)
    - Estimated usage should be calculated dynamically (by changing the quantity, the usage should start tracking for that particular day)
- Responsive web design because it should be used as a mobile web app
- Notification bar to give a quick TODO on what items are going to be out of stock soon from the dashboard instead of having to click into each item
- Actionable sustainable procurement alternatives:
    - Requires data entry into existing suppliers and alternative suppliers within the local area
    - Likely use AI to consider the